

ONATCHIWAY, QC, Canada

WMO: 713870

Lat: 48.894N

Lon: 71.032W

Elev: 304

StdP: 97.73

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.1	-30.7	-38.2	0.1	-34.0	-34.6	0.1	-30.5	7.6	-8.4	6.9	-9.7	0.7	0	0.596

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	27.9	18.4	26.2	17.7	24.5	17.0	20.7	25.0	19.6	23.4	18.7	22.2	2.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.2	14.5	22.2	18.3	13.7	21.5	17.3	12.8	20.6	61.2	25.0	57.6	23.3	54.3	22.1	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.2	6.2	5.3	-38.3	31.1	3.3	1.7	-40.6	32.3	-42.5	33.3	-44.4	34.3	-46.8	35.5		
			-38.4	22.5	3.4	1.1	-40.8	23.3	-42.8	23.9	-44.7	24.5	-47.1	25.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.0	-15.9	-14.1	-7.6	0.9	9.2	14.1	16.8	15.8	11.4	4.7	-2.1	-10.7
	DBStd	12.70	8.31	7.62	7.43	4.98	4.52	4.14	3.13	3.28	4.36	4.26	5.41	7.23
	HDD10.0	3589	804	676	544	273	71	9	0	1	33	172	362	643
	HDD18.3	6029	1062	909	803	523	286	138	68	93	212	422	612	902
	CDD10.0	656	0	0	0	1	46	133	211	180	76	9	1	0
	CDD18.3	53	0	0	0	0	2	12	21	14	4	0	0	0
	CDH23.3	612	0	0	0	0	60	178	224	121	28	1	0	0
	CDH26.7	113	0	0	0	0	12	44	42	13	3	0	0	0
Wind	WSAvg	2.1	2.0	2.1	2.5	2.7	2.3	2.1	1.8	1.8	1.9	2.2	2.3	2.1
Precipitation	PrecAvg	1016	64	56	55	63	80	96	125	109	111	97	79	80
	PrecMax	1175	145	111	111	106	158	149	292	242	221	145	127	118
	PrecMin	881	22	30	21	29	31	47	50	51	58	37	26	40
	PrecStd	81	24	20	24	23	28	27	46	41	33	28	21	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.9	6.1	12.3	17.7	28.2	30.6	30.1	28.2	26.5	20.1	13.8	6.7
		MCWB	5.1	4.1	5.5	8.7	14.9	19.8	19.7	19.7	20.0	13.9	11.4	5.6
	2%	DB	2.2	2.2	7.7	14.0	24.7	27.7	27.7	26.3	23.5	16.9	9.9	3.1
		MCWB	1.6	0.5	3.5	6.5	13.4	17.7	18.8	18.7	18.3	13.2	8.0	2.3
	5%	DB	-0.3	0.3	5.1	11.4	21.9	25.2	25.9	24.6	21.1	14.4	7.4	1.2
		MCWB	-1.0	-0.9	1.6	5.7	12.7	16.2	18.1	18.0	16.8	10.9	5.8	0.4
	10%	DB	-3.5	-1.7	2.9	8.9	18.9	22.9	24.1	22.7	19.1	11.8	5.0	-0.4
		MCWB	-4.2	-2.8	0.8	4.4	11.8	15.5	17.5	17.2	15.6	9.3	3.8	-1.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.2	4.4	7.2	11.2	18.2	21.7	22.4	21.4	21.0	15.9	11.4	5.6
		MCDB	5.8	5.3	8.9	15.6	22.9	28.2	26.9	25.8	25.4	18.7	13.2	6.3
	2%	WB	1.8	0.7	4.0	8.3	15.9	19.7	20.7	20.2	19.3	13.5	8.5	2.4
		MCDB	2.2	1.8	6.5	12.0	21.8	23.7	24.6	24.2	22.2	16.6	9.6	3.0
	5%	WB	-0.9	-0.8	2.4	6.3	13.9	18.2	19.5	19.2	17.7	11.7	6.1	0.5
		MCDB	-0.4	0.4	4.7	10.3	19.6	22.8	23.5	22.5	20.5	13.9	7.1	1.1
	10%	WB	-4.3	-2.9	0.7	4.6	12.3	16.8	18.5	18.3	16.2	9.7	4.1	-1.1
		MCDB	-3.7	-1.7	2.7	8.3	18.2	21.2	22.2	21.5	18.8	11.7	5.1	-0.5
Mean Daily Temperature Range	5% DB	MDBR	12.3	13.7	13.5	11.5	13.6	13.3	12.1	11.7	11.2	8.3	7.2	9.5
		MCDBR	11.2	11.9	14.9	16.3	20.4	17.6	15.7	14.7	13.9	13.7	9.8	8.9
		MCWBR	10.9	10.4	10.8	10.1	10.4	8.7	7.9	7.9	8.7	9.5	8.1	8.4
	5% WB	MCDBR	11.5	10.9	12.0	14.0	15.1	13.8	11.7	11.2	11.6	11.8	8.8	8.8
		MCWBR	11.4	9.9	9.1	9.6	8.8	7.7	6.5	6.5	7.8	9.0	7.9	8.7
Clear-Sky Solar Irradiance	taub	0.257	0.276	0.302	0.365	0.347	0.375	0.390	0.372	0.359	0.320	0.303	0.255	
	taud	2.281	2.234	2.211	2.222	2.438	2.385	2.361	2.414	2.434	2.531	2.418	2.338	
	Ebn at Noon	811	876	909	880	913	885	865	866	842	827	750	766	
	Edh at Noon	71	95	116	129	108	115	117	106	94	72	64	59	
All-Sky Solar Radiation	RadAvg	1.25	2.25	3.51	4.49	5.19	5.52	5.14	4.32	3.29	1.88	1.01	0.80	
	RadStd	0.17	0.29	0.32	0.36	0.49	0.39	0.48	0.27	0.25	0.20	0.10	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	-0.76	N/A	N/A	N/A	N/A	N/A	+246	+262	N/A	N/A
(47)	Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page